

10/02/2026: Midterm Review Part 1: Proofs

CSCI 246: Discrete Structures

October 2, 2026

Announcements

At beginning of class Monday, you can optionally retake any quiz of **your choice!**

The study guides will be the same. The questions will be different (in general).

Today's Agenda

- Quiz (20 mins)
- Review of proof techniques (5-10 mins)
- Group exercises (20-25 mins)

Motivational point for this age of AI

Finding: A person's computer science skills, cognitive ability, and writing ability are strong predictors of the quality of software produced when using LLM coding agents.

Computer Science Achievement and Writing Skills Predict Vibe Coding Proficiency

Sverrir Thorgeirsson*
ETH Zürich
Switzerland
sverrir.thorgeirsson@inf.ethz.ch

Theo B. Weidmann*
ETH Zürich
Switzerland
theo.weidmann@inf.ethz.ch

Zhendong Su
ETH Zürich
Switzerland
zhendong.su@inf.ethz.ch

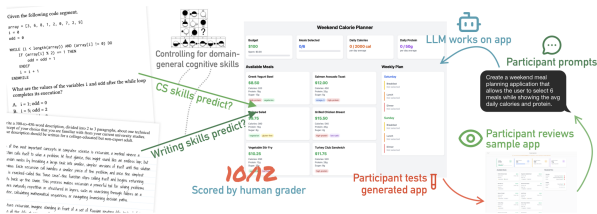


Figure 1: Our study investigates how writing skills and CS achievement relate to the ability to vibe code when controlling for domain-general cognitive skills. Participants reviewed sample applications, drafted prompts to an LLM-based agent, tested the resulting applications, and made further refinements. The final applications were then scored by a human grader.

Weekly Quiz #5

1. (Partitions.) How many different anagrams (including nonsensical words) can be made from the word bookkeeper? Justify your answer.
2. (Relations.) Consider the relation $\leq$ (less than or equal to) on the integers. Is the relation reflexive? Symmetric? Transitive? Justify your answers.
3. (Equivalence relations.) Let n be a positive integer. Prove that congruence modulo n is an equivalence relation on the set of integers.

Reference Material: Definition

Let n be a positive integer. We say that integers x and y are **congruent modulo n** , and we write

$$x \equiv y \pmod{n}$$

if $n|(x - y)$.

Review of proofs

Proof Template: Direct proof of an if-then (if A then B) theorem

1. Write down the if-then statement you're trying to prove.
2. At the beginning of the proof, write down the *antecedent* (the A in if A then B) as your assumption.
3. At the end of the proof, write down the *consequent* (the B in if A then B) as your conclusion.
4. Unravel the definitions, working forward from the beginning of the proof and backward from the end of the proof.
5. Forge a link between the two halves of the argument.

Example

Proposition. The square of an odd integer is odd.

Proof.

Annotation	Main Text
Convert Prop. to “if-then” form	We show that if x is an odd integer, then x^2 is odd.
State antecedent	Let x be an odd integer.
Unravel defs.	Then by definition of <i>odd</i> , there is an integer a such that $x = 2a + 1$.
*** The glue ***	So $x^2 = (2a + 1)(2a + 1) = 4a^2 + 4a + 1 = 2(2a^2 + 2a) + 1$.
Unravel defs.	So there is an integer b (where $b = 2a^2 + 2a$) such that $x^2 = 2b + 1$.
State consequent	Hence, x^2 is odd.

Proof Template: Direct proof of an if-and-only-if theorem

To prove a statement of the form “A iff B”:

- ($\implies$) Prove “If A, then B”.
- ($\impliedby$) Prove “If B, then A”.

Scheinerman Proof Template 2

Proof Template: Refuting a false if-then statement via a counterexample

To disprove a statement of the form “If A then B ”:
Find an instance where A is true but B is false.

Scheinerman Proof Template 3

Proof Template: Refuting a false if-then statement via a counterexample

To disprove a statement of the form “If A then B ”:
Find an instance where A is true but B is false.

Scheinerman Proof Template 3

Note. Often, you'll need to **justify** that A is true (by showing that the conditions of the relevant definition is satisfied.)

Proof Template: Proving one set is a subset of another

To show $A \subseteq B$:

Show that if $x \in A$, then $x \in B$.

Scheinerman Proof Template 6

Proof Template: Proving two sets are equal

Let A and B be sets. To show $A = B$:

- Show $A \subseteq B$.
- Show $B \subseteq A$.

Scheinerman Proof Template 6

Proof Template: Proving two sets are equal

Let A and B be sets. To show $A = B$:

- Show $A \subseteq B$.
- Show $B \subseteq A$.

Scheinerman Proof Template 6

Remark. In other words, we must show:

- If $x \in A$, then $x \in B$.
- If $x \in B$, then $x \in A$.

Group exercises

Ashley, Isabelle: 7	Hager, Nathan: 11	Putnam, John: 5
Bailey, Tanna: 7	Horner, Jayden: 7	Ramos, Vydel: 10
Blaisdell, Olyver: 5	Johnson, Jaret: 3	Rathbun, Jedidiah: 9
Brown, Ashton: 1	Kintzel, Samantha: 1	Roberts, Matthew: 13
Brown, Jonathan: 2	Lameres, Alexis: 9	Roller, Nathan: 4
Cardenas, Dominic: 2	Lee, Holden: 5	Ryan, Otis: 3
Cutler, George: 11	Mayala, Reece: 6	Shevtsov, Timothy: 1
Fabik, Roland: 8	McLean, Connor: 12	Sinclair, Aidan: 11
Fellows, Audrey: 10	Mcbride, Daniel: 10	Smith, Charles: 6
Fitzgerald, Ryan: 8	Miller, Maverick: 2	Stensland, Emma: 13
Fuller, Ryan: 6	Moss, Emily: 8	Thiessen, Logan: 9
Griffen, Clara: 4	Mousseau, Matthew: 3	Thompson, Jaiden: 4
Grutsch, Tanner: 12	Neuman, Evan: 13	Vath, Ethan: 12

Group exercises

1. Prove that the product of two even integers is even.
2. Let x be an integer. Prove that $0 \mid x$ if and only if $x = 0$.
3. Disprove: If p and q are prime, then $p + q$ is composite.
4. Let $A = \{x \in \mathbb{Z} : 4 \mid x\}$ and let $B = \{x \in \mathbb{Z} : 2 \mid x\}$. Show that $A \subseteq B$.
5. Let $A = \{n \in \mathbb{Z} : 3 \mid n\}$ and let $B = \{n \in \mathbb{Z} : 3 \mid (n + 6)\}$. Show that $A = B$.

Reference: Definitions

- Let a, b be integers. We say that a is **divisible** by b , written $b \mid a$, if there is an integer c such that $bc = a$. We call b a **divisor** of a .
- An integer is called **even** if it is divisible by 2. (That is, an integer x is even if there is an integer c such that $x = 2c$.)
- An integer p is called **prime** if $p > 1$ and the only positive divisors of p are 1 and p .
- A positive integer a is called **composite** if there is an integer b such that $1 < b < a$ and $b \mid a$.

Solution: Group exercise #1

Problem. Prove that the product of two even integers is even.

Solution. We show that if $z = xy$ where x and y are even, then z is even. Let x and y be even. Then $x = 2a$ where a is an integer and $y = 2b$ where b is an integer. So $z = xy = (2a)(2b) = (4ab) = 2(2ab)$. Hence $z = 2c$ for some integer c (namely $c = 2ab$). Hence z is even.

Reference: Definition

An integer is called **even** if it is divisible by 2. (That is, an integer z is even if there is an integer c such that $z = 2c$.)

Solution: Group exercise #2

Problem. Let x be an integer. Prove that $0 \mid x$ if and only if $x = 0$.

Solution. First we show that if $0 \mid x$ then $x = 0$. Let $0 \mid x$. Then there is an integer c such that $0c = x$. So $x = 0$.

Now we show that if $x = 0$ then $0 \mid x$. Let $x = 0$. Let c be any integer (e.g. $c = 7$). Then $x = 0c$. Hence, there is an integer c such that $0c = x$. Hence $0 \mid x$.

Reference: Definition

Let a, b be integers. We say that a is **divisible** by b , written $b \mid a$, if there is an integer c such that $bc = a$.

Solution: Group exercise #3

Problem. Disprove: If p and q are prime, then $p + q$ is composite.

Solution. We give a counterexample where p and q are prime, but $p + q$ is not composite. Let $p = 2$ and $q = 3$. Then p and q are both prime. ($2 > 1$ and the only positive divisors of 2 are 1 and 2, and $3 > 1$ and the only positive divisors of 3 are 1 and 3.). However, $p + q = 5$, which is prime ($5 > 1$ and the only positive divisors of 5 are 1 and 5), and therefore not composite.

Reference: Definitions

- An integer p is called **prime** if $p > 1$ and the only positive divisors of p are 1 and p .
- A positive integer a is called **composite** if there is an integer b such that $1 < b < a$ and $b \mid a$.

Solution: Group exercise #4

Problem. Let $A = \{x \in \mathbb{Z} : 4 \mid x\}$ and let $B = \{x \in \mathbb{Z} : 2 \mid x\}$. Show that $A \subseteq B$.

Solution. We will show that if $x \in A$, then $x \in B$. Let $x \in A$. Then $4 \mid x$. So there is an integer c such that $x = 4c$. So $x = 2(2c)$. So $x = 2d$ for some integer d (namely $d = 2c$). So $2 \mid x$. So $x \in B$.

Reference: Definition

Let a, b be integers. We say that a is **divisible** by b , written $b \mid a$, if there is an integer c such that $bc = a$. We call b a **divisor** of a .

Solution: Group exercise #5

Problem. Let $A = \{n \in \mathbb{Z} : 3 \mid n\}$ and let $B = \{n \in \mathbb{Z} : 3 \mid (n + 6)\}$. Show that $A = B$.

Solution. First we show that if $x \in A$, then $x \in B$. Let $x \in A$. Then $3 \mid x$. So $x = 3c$ for some integer c . So $x + 6 = 3c + 6$. So $x + 6 = 3(c + 2)$. So $x + 6 = 3d$ for some integer d , namely $d = c + 2$. So $3 \mid (x + 6)$. So $x \in B$.

Now we show that if $x \in B$, then $x \in A$. Let $x \in B$. So $3 \mid (x + 6)$. So $3c = x + 6$ for some integer c . So $3c - 6 = x$. So $3(c - 2) = x$. So $3d = x$ for some integer d (namely $d = c - 2$). So $x \in A$.

Reference: Definition

Let a, b be integers. We say that a is **divisible** by b , written $b \mid a$, if there is an integer c such that $bc = a$. We call b a **divisor** of a .